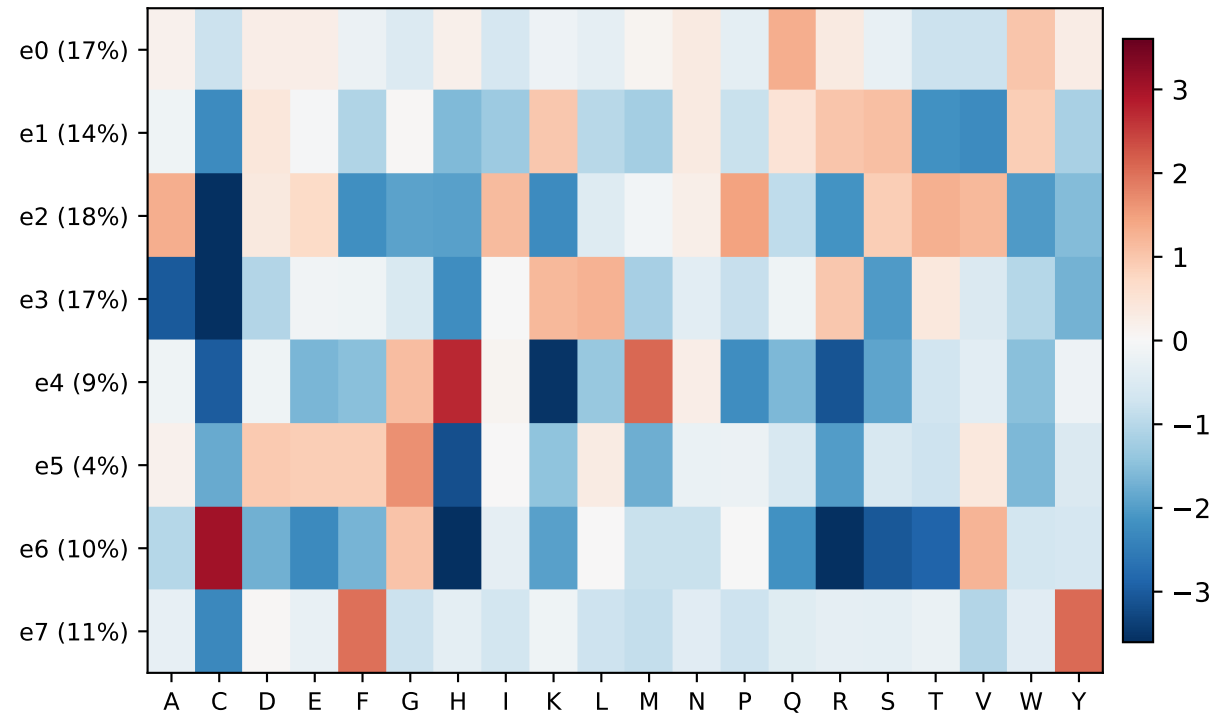
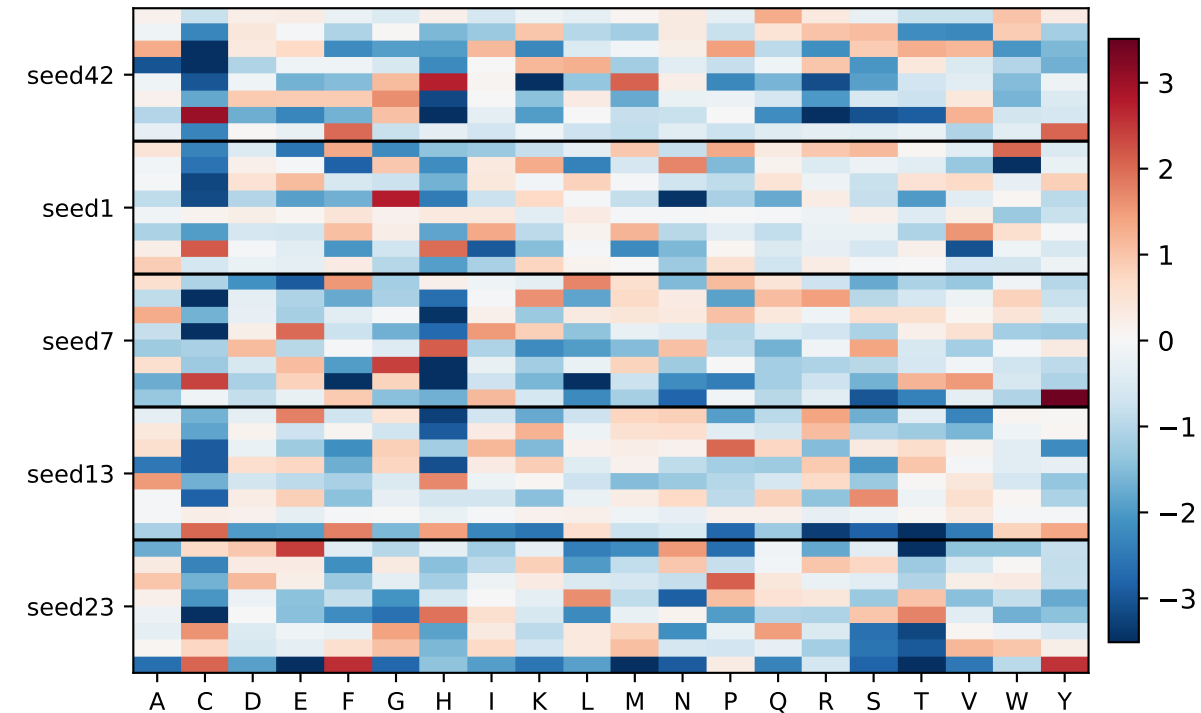


v24 per-residue MoE: what each expert reads on the DBD (2260 unique DBDs, 323,751 routed residues)

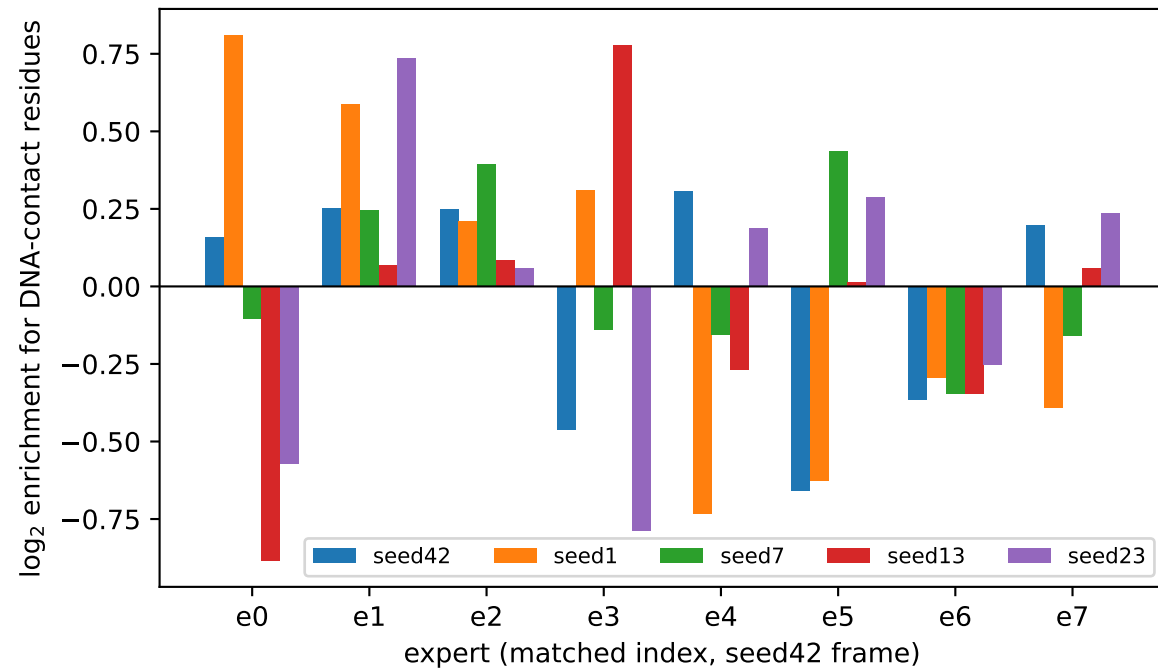
A seed42: expert x residue enrichment
 $\log_2 P(\text{aa}|\text{e}) / E_{\text{family}}[P(\text{aa})]$ (family composition controlled)



B all 5 ensemble members, experts Hungarian-matched onto seed42
 (shared vertical structure = reproducible archetype)



C do experts concentrate co-crystal base-contacting residues?
 background contact rate = 0.1951



D are expert archetypes reproducible across seeds?

